

Test Plan - KaloriN AI

Document Information

Item	Details
Test Plan ID	TP-KALORIN-001
Version	1.0
Project Name	KaloriN AI
Application Type	Nutrition Tracking Web Application
Prepared By	Naufal Fadhiil
Role	QA Engineer
Date Created	June 2026
Testing Type	Manual Testing

1. Introduction

A. Background

KaloriN AI is a nutrition tracking web application that helps users monitor their nutritional intake through food analysis, AI-powered food scanning, meal tracking, personalized recommendations, BMI calculation, calorie targets, and weekly nutrition insights.

The application combines traditional web functionalities with machine learning services to provide personalized nutrition experiences.

B. Purpose

This Test Plan defines the testing scope, objectives, strategy, resources, deliverables, and quality criteria used during the testing activities of KaloriN AI.

The purpose of testing is to ensure that the application's core functionalities operate correctly, user workflows function as intended, and identified defects are documented and verified before project completion.

2. Test Objectives

The objectives of testing are:

1. Verify that all core features function according to requirements.
2. Validate business rules and nutrition calculations.
3. Verify authentication and access control mechanisms.
4. Ensure user workflows operate correctly from onboarding to nutrition tracking.
5. Validate responsiveness across different screen sizes.
6. Identify and document defects.
7. Verify defect fixes through retesting and regression testing.
8. Assess AI Scanner functionality from an end-user perspective.

3. Test Items

No	Module Name
1.	Authentication
2.	Protected Routes
3.	User Onboarding
4.	Home
5.	Analyze
6.	Meals
7.	Track
8.	Insights
9.	Profile Management
10.	Navigation & Sidebar
11.	Error Handling
12.	Responsiveness

4. Test Items

A. Features Included

1. Authentication
 - Registration
 - Login
 - Logout
 - Session Handling
2. User Onboarding
 - Personal Information
 - Height Input
 - Weight Input
 - Activity Level
 - Goal Selection
3. Analyze
 - Food Search
 - AI Food Scanner
 - Food Information Display
4. Meals
 - Personalized Recommendations
 - Recommendation Details
5. Track
 - Meal Logging
 - Nutrition Tracking
 - Daily Progress
6. Insights
 - Weekly Summary
 - Nutrition Trends
 - AI Insights
7. Profile
 - Profile Update
 - Goal Management
 - Body Statistics
8. Cross Functional Areas
 - Navigation
 - Responsive Design
 - Error Handling

B. Out of Scope

- Security Testing
- Penetration Testing
- Performance Testing
- Load Testing
- Stress Testing
- Accessibility Testing
- Source Code Review
- AI Model Training Validation

5. Test Strategy

A. Testing Approach

Testing was performed using a Black Box Testing approach. The application was tested from an end-user perspective without reviewing internal implementation details.

Testing focused on:

- User workflows
- Functional behavior
- Business logic
- User interface consistency
- Responsive behavior

B. Testing Technique

Technique	Purpose
Equivalence Partitioning	Valid and invalid input testing
Boundary Value Analysis	Input limit validation
Use Case Testing	End-to-end workflow validation
Exploratory Testing	Discovery of unexpected defects

Regression Testing	Verification after bug fixes
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C. Testing Type

Testing Type	Applied
Functional Testing	Yes
UI Testing	Yes
Validation Testing	Yes
Exploratory Testing	Yes
Responsive Testing	Yes
Regression Testing	Yes
Cross Browser Testing	Yes
Security Testing	No
Performance Testing	No

6. Test Environment

A. Device

- Laptop / Desktop PC
- Stable Internet Connection

B. Software

Item	Version
Chrome	Latest
Edge	Latest

C. Test Management Tools

Tool	Purpose
Qase	Test Case & Test Scenario Management
GitHub	Project Repository
Postman	API Testing
Google Docs / Spreadsheet	Documentation

7. Entry Criteria

Testing activities may begin when:

- Application deployment is available
- Required modules are accessible
- Test environment is operational
- Test environment is operational
- Test data is prepared

8. Exit Criteria

Testing activities are considered complete when:

- All planned test cases have been executed
- Critical defects have been addressed
- Retesting has been completed
- Regression testing has been completed
- Test summary report has been finalized
- QA sign-off has been completed

9. Test Deliverables

No	Deliverable	Description
1.	Feature Inventory	A list of application modules and features used to define testing scope and prioritize testing activities. This serves as the foundation for all subsequent testing documentation.
2.	Test Plan	A document describing testing objectives, scope, strategy, resources, risks, and overall testing approach for the project.
3.	Test Cases (Qase)	Detailed test cases containing test steps, test data, and expected results used during execution.
4.	Test Execution Result	The execution outcome of all test cases, including Pass, Fail, and Skipped statuses recorded in Qase.
5.	Bug Report	Documentation of defects discovered during testing, including defect description, severity, reproduction steps, and supporting evidence.
6.	Retest Result	Verification results after developers fixed reported defects to confirm that issues were successfully resolved.
7.	Regression Verification	Additional testing performed after bug fixes to ensure existing functionalities were not negatively impacted.
8.	Test Summary Report	A high-level summary of testing activities, execution metrics, defect statistics, risks, findings, and recommendations.
9.	Final Test Plan Archive	Finalized testing documentation package maintained as project reference and portfolio evidence.

10. Risks and Mitigation

Risk	Mitigation
Tight project deadline	Prioritize critical user flows

Limited testing resources	Risk-based testing approach
AI Scanner accuracy variation	Document observations and recommendations
Browser compatibility issues	Perform cross-browser testing

11. AI Feature Consideration

KaloriN AI includes an AI-powered food scanning feature.

Testing focused on verifying:

- Scanner accessibility
- Upload functionality
- Result generation
- User-facing output

The accuracy of machine learning predictions and model training quality were considered outside the scope of software functional testing.

However, observations regarding food recognition quality were documented within the Test Summary Report as product improvement opportunities.

Approval

Role	Name	Date
QA Engineer	Naufal Fadhiil	03 June 2026